

# Bridge Note — Product Definition

**Product:** Bridge Note — an AI-assisted writing companion that helps a young person in Libya take the hardest step in asking for help: writing the first message to a real person.

Event: Ai4LY National Codathon — Libya Artificial Intelligence Forum 2026

Document type: Product Definition (the foundation for implementation)

Team: four members

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Status: Product level only. Technical decisions are deliberately left open and are made after the validation work described in Part IV.

## How to read this document

Every section sits on one conceptual chain. Each major section states where it sits on that chain.S1 Problem → S2 User → S3 Interaction → S4 Observation → S5 Private on-device record

→ S6 Writing trigger → S7 Writing experience → S8 Intended outcome

This is a chain of product behaviour, not a technical architecture.

Source tags used throughout show where a requirement comes from:

| Tag    | Source                                                                                       |
|--------|----------------------------------------------------------------------------------------------|
| [L§n]  | AI Mental Health Challenge — Lecture / Meeting Summary, section n                            |
| [T§n]  | Ai4LY National Codathon — Participation Acceptance Notice, Terms and Requirements, section n |
| [TP]   | The team's Codathon Project Scope and Idea Selection Plan                                    |
| [Team] | A product decision made by this team, with the reason stated                                 |

**Labels used for uncertainty.** Where the project material does not answer something, it is marked as one of: **Unresolved question**, **Assumption**, **Decision requiring validation**, or **Needs research**. Nothing unknown has been given an invented answer.

# PART I — THE PRODUCT

## 1. Purpose of This Document

This document gives the four-person team one shared, complete understanding of what we are building, why, how it behaves as a connected system, what it must accomplish, what constrains us, which implementation choices remain open, and what must be validated before we commit to them.

It answers, in order:

- What exactly are we building, and why must it exist?
- Who is it for, and what does the user actually do?
- What does the product do in response, what does it record, and how is that kept safe?
- How does the user's interaction lead to the writing experience, and what outcome are we after?
- What must it accomplish, and what are our real constraints?
- What do we still need to decide, research and confirm?

**Scope of the document.** It defines WHAT and WHY. It does not define HOW in engineering terms. A product *requirement* says what the product must accomplish. An implementation choice says how we might accomplish it. The two are kept apart throughout so a technology never silently becomes a requirement.

**Key terms.**

| Term | Meaning in this product                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Chip | A one-tap choice describing a situation the user is dealing with (for example exams, family, work, relationships, sleep, money, other). |
| Note | The short message the user writes and sends to a person of their choice.                                                                |

| Term                   | Meaning in this product                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Private record</b>  | A small, user-controlled, on-device memory of the user's own recent chip selections. It exists only to serve the user's own experience. |
| <b>Writing trigger</b> | A gentle, optional question that invites the user to begin writing a note.                                                              |
| <b>Draft</b>           | A tone-adapted version of the note proposed by the product. The user edits it.                                                          |
| <b>Guardian Layer</b>  | The safety wrapper that runs before any drafting and keeps a route to a human always available.                                         |
| <b>Human route</b>     | A verified way to reach a real person: a trusted individual the user chooses, or a verified support contact.                            |

## 2. Why This Product Exists

*Place on the chain: **S1 Problem**.*

### 2.1 The Issue

The hardest step in a mental health struggle is often the first sentence. A young person in Libya who wants to reach out to a friend, a sibling or a counsellor faces a blank page at exactly the moment they have the least capacity to fill it. Three forces compound:

- **Stigma.** Mental illness has long been treated as a private family matter, so naming a feeling out loud carries social cost. Neutral framing matters more than eloquence.
- **Emotional overwhelm.** The person knows *something* is wrong but cannot order it into a sendable message.
- **Articulation cost.** Turning "I feel bad and exams are destroying me" into something a friend can act on takes energy the person does not have.

The result is silence, and silence is the failure mode. Early signs go unshared until they are no longer early. The lecture's own examples of stress — academic pressure, work

pressure, family and relationship pressure [L§5] — are exactly the pressures that produce this silence in the target age group.

**Design consequence:** the product's job is not to hold the conversation. Its job is to end the silence early and hand the user to a human.

## 2.2 The Problem Statements

| ID | Problem                                                                                                                                                              | Source      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| P1 | Young people stay silent at the first sentence of asking for help, because of stigma, overwhelm and the effort of articulating.                                      | [L§1, L§5]  |
| P2 | Early signs of stress go unnoticed and unshared, so help is sought late rather than early.                                                                           | [L§1, L§6]  |
| P3 | Specialist support is scarce and often avoided, so the people most able to help early are usually already in the young person's life — if the person can reach them. | [L§2, L§12] |
| P4 | Any tool in this space carries risk: it can expose sensitive data, give harmful guidance, imply a diagnosis, or start to replace a human.                            | [L§3]       |

## 2.3 The Motive

Break the silence by removing the friction of starting the conversation.

Bridge Note is **not** a conversational chatbot, **not** an artificial therapist, and **not** a tool designed to keep the user engaged with a screen. Every design decision is measured against one question: Did the user reach a real human?

Faster is better than longer. Shorter sessions are better than deeper ones. An app that gets closed because the user is now talking to their sister has succeeded. This fits the challenge's stance that the solution may assist, guide or provide early support but must not replace specialists [L§3], and the safer general pattern for AI in this area: fast hand-off to human care rather than extended AI conversation.

## 3. Who It Is For

*Place on the chain: **S2 User**.*

**Primary user.** A young person in Libya (a student or young adult, some of them under 18) who is under everyday pressure — exams, work, family, relationships, sleep, money — and who would benefit from telling someone, but cannot start.

**Secondary person.** The recipient: a friend, sibling, parent, trusted adult, counsellor or teacher whom the user chooses. The recipient never uses the product; they receive an ordinary message from someone they know.

### 3.1 What the user needs

| ID | User need                                                                                                           | Addresses | Source        |
|----|---------------------------------------------------------------------------------------------------------------------|-----------|---------------|
| N1 | Help starting: something that makes the first sentence possible, even with almost no energy or typing.              | P1        | [L§1, L§5] ▾  |
| N2 | Neutral, non-clinical wording, so nothing they share reads as a label they attached to themselves.                  | P1        | [L§3] ▾       |
| N3 | Privacy and anonymity: no account, no identification, nothing kept that they did not choose to keep.                | P4        | [L§3] ▾       |
| N4 | Control: nothing is sent, and nobody is contacted, unless the user chooses; the user chooses who receives the note. | P4        | [L§3, Team] ▾ |

| ID | User need                                                                                                                                    | Addresses | Source        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|
| N5 | A reliable route to a human when things are serious, without depending on the product to judge correctly.                                    | P3, P4    | [L§1, L§3] -  |
| N6 | To be gently noticed: a quiet recognition that the same pressure keeps returning, with an easy way to act on it.                             | P2        | [L§6] -       |
| N7 | A product that works in their language and conditions: Arabic, Libyan dialect and mixed-script writing, patchy connections, ordinary phones. | P1, P3    | [L§4, L§12] - |

### 3.2 Assumptions about the user

- **Assumption:** the user has a phone and already uses at least one messaging app.
- **Assumption:** the user can write one to three short lines, in Arabic, Libyan dialect, mixed Arabic/English or Arabic written in Latin letters.
- **Unresolved question:** how a minor (under 18) uses the product, and what the product must say or ask about age. See Q5.

## 4. The Product Idea

Place on the chain: **S1 → S8, whole system.**

### 4.1 Overview

Bridge Note is a **local-first** product that turns a user's own words into a message they are willing to send.

The user taps situation chips, writes one to three lines in their own words, picks a recipient, and reads three tone-adapted drafts (gentle, direct, formal). They edit a draft and share it through a messaging app they already use. The AI never sends anything and never contacts anyone on the user's behalf.

A small private record of the user's own chip selections can be kept on their device. When the same pressure keeps coming up, that record lets the product ask one quiet, appropriate question — *"You've mentioned exams a lot lately. Do you want help writing a short note to a friend about it?"* — which invites the user to begin writing. The diary-like element is therefore not a generic notes feature; it is the memory that connects what the user taps to the moment they are invited to write.

One sentence. Bridge Note removes the hardest step in seeking help — writing the first sentence — by turning a few chips and one to three lines of the user's own words into an editable, tone-adapted message, while a safety layer guarantees the user is routed to a verified human and is never diagnosed, never profiled and never left with data behind.

## 4.2 Product principles

1. **Human first.** Every flow ends at a person, not at the product. [Team]
2. **Own words.** The user remains the author; the product proposes, the user approves. [Team]
3. **Private by default.** The least information needed, kept in the fewest places, under the user's control. [L§3]
4. **Safe before helpful.** Risk is checked before any drafting happens. [L§3]
5. **Useful with nothing.** The product works with zero history and zero typing, using chips alone. [Team]
6. **Honest about limits.** It says plainly what it is not, and measures its safety claims rather than asserting them. [L§3, L§12]

## 4.3 The four layers of the product

### Layer 1 — Capture (deterministic, no AI).

Situation chips (exams / family / work / relationships / sleep / money / other), a short free-text box (one to three lines), and a recipient selector (friend, sibling, parent, trusted adult, counsellor or teacher). The chip set means the product is useful even if the user writes nothing at all.

### Layer 2 — Drafting Engine (the AI core).

The AI restructures the user's free text and produces three drafts at three tones. Hard constraints on generation: no condition names, no diagnostic language, no medication or treatment suggestions, no invented facts. Every generated claim must trace to something the user actually typed.

### Layer 3 — Guardian Layer (safety, mandatory).

A risk check runs before any drafting. On a trigger the product skips drafting, shows a

fixed support card first, and offers to continue with the note afterwards. Defined in section 10.4.

#### **Layer 4 — Trust and Control.**

Capabilities that let the user see and steer what the AI does: a Visible Faithfulness Map that highlights which generated phrases map to which input phrases, so the user can see the AI did not invent their feelings; a **Live Baseline Toggle** that compares the AI draft against a plain template so the user, and a judge, can see whether the AI actually helped; a view of exactly what would leave the device before anything is sent; and a persistent "Talk to Someone Now" control present on every screen that never waits for an algorithmic trigger.

## **4.4 What the AI actually does**

The lecture stressed that labelling an idea "AI-based" is not enough [L§4]. This is the precise division of labour.

#### **AI does four things that rules, forms or templates cannot do:**

1. **Structures messy free text.** The user writes: *"i cant do this anymore with exams and my dad keeps asking and i just freeze"*. The AI separates the situation (exams), the impact (freezing, pressure from a parent) and the implicit need (to be left alone, or to be helped) into a sendable structure. A form cannot do this because the user will not fill in a form in this state; a rule set cannot, because the input is unstructured natural language.
2. **Adapts tone for an audience.** The same content must read differently to a best friend, a mother and a school counsellor. This is a genuine language-generation task with a social outcome, not string substitution.
3. **Verifies its own faithfulness.** The Faithfulness Map aligns generated phrases back to the user's input — the AI checking AI, shown to the user, so the model's role is demonstrable rather than decorative.
4. **Classifies risk.** The risk check applies model judgement to Arabic natural language where a phrase list alone produces misses. This is the highest-stakes AI function in the product.

**What the AI explicitly does not do:** it does not diagnose, suggest medication or treatment, converse therapeutically, decide who to contact, send anything, or retain anything after the session.

The Live Baseline Toggle is the proof. It lets anyone see on screen the difference between the AI draft and a plain template. If the AI output is not meaningfully better, the toggle shows that too — which is why it exists rather than being hidden.

**What the deterministic parts do.** The chip capture, the private record, the choice of which gentle question to ask, and the fixed support card are deliberately not AI. Predictable behaviour is the right tool where the stakes are highest and where the answer is simple.

## 5. The Overall Product Flow

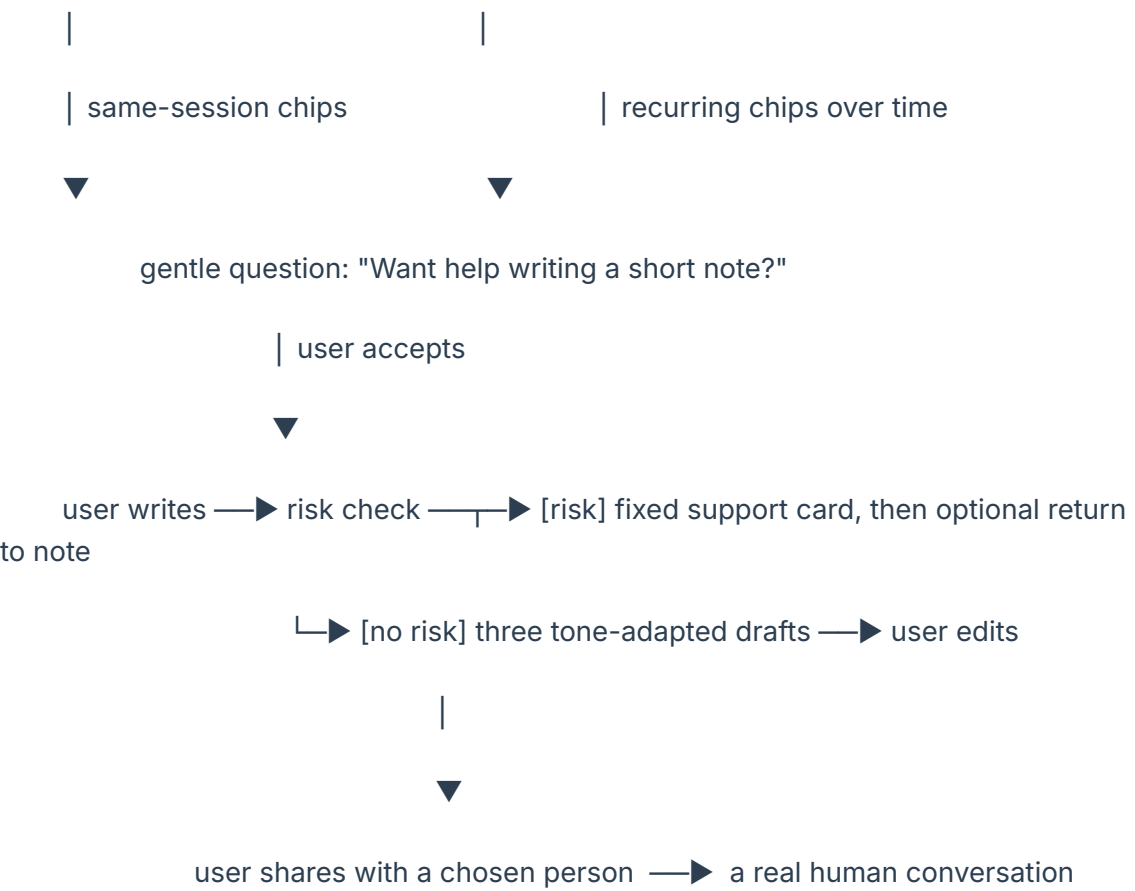
*This section states the whole chain. Later sections each develop one stage.*

| Stage                              | What happens                                                                                                                                   | Product section |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>S1 Problem</b>                  | Silence at the first sentence; early signs unshared.                                                                                           | §2              |
| <b>S2 User</b>                     | A young person under everyday pressure, who needs a way to start.                                                                              | §3              |
| <b>S3 Interaction</b>              | The user taps chips (often, quickly) and may write a few lines and pick a recipient.                                                           | §6              |
| <b>S4 Observation</b>              | The product receives only what the user deliberately gives it inside the product: chip selections, optional text, recipient and tone choices.  | §7              |
| <b>S5 Private on-device record</b> | Relevant selections can be remembered privately on the device, under the user's control, never transmitted.                                    | §8              |
| <b>S6 Writing trigger</b>          | The product can ask one gentle question — now, from this session's chips, or over time, from recurring chips — that invites the user to write. | §9              |
| <b>S7 Writing experience</b>       | The user writes in their own words; risk is checked first;                                                                                     | §10             |

| Stage                      | What happens                                                                                         | Product section |
|----------------------------|------------------------------------------------------------------------------------------------------|-----------------|
|                            | three drafts appear; the user edits; the user decides whether and how to share.                      |                 |
| <b>S8 Intended outcome</b> | The user sends a message to a real person they chose, and knows where to turn if it is more serious. | §11             |

**Two ways into the writing experience.** The user can open the product and start writing at any time (no trigger needed). The trigger is an additional, optional invitation for moments when the user's own interactions show a recurring pressure.

One way out, always available. At every stage, the persistent human route is reachable. A risk detection at S7 redirects to it. The user never needs the product to be right in order to reach a person. user taps chips —> product receives selections —> (if user allows) private on-device record



# PART II — HOW THE PRODUCT WORKS

## 6. The User's Interaction

*Place on the chain: **S3 Interaction**. Serves N1, N2, N4, N7. Addresses P1.*

The interaction is built to cost almost nothing, because the user starts with almost nothing.

**What the user does, in the order they can do it:**

1. **Taps one or more chips.** This alone is a complete, useful act. It tells the product what kind of pressure the user is under, without a word typed. Chips are frequent, quick, and the main way the product learns how the user is using it. [Team]
2. **Optionally writes one to three lines** in their own words, in whatever language or mixture feels natural (Arabic, Libyan dialect, mixed Arabic/English, Arabic in Latin letters). [L§12, Team]
3. **Chooses who the note is for:** friend, sibling, parent, trusted adult, or counsellor/teacher. Nothing is imposed; the user picks. The recipient options include family and a trusted adult so the product respects the user's own values and relationships. [Team]
4. **Reads three drafts** (gentle, direct, formal), edits one, and decides whether to share.
5. **Shares through a messaging app they already use**, or does not share.

**Behaviour requirements for the interaction:**

- Usable with zero typing (chips only) and better with up to three lines.
- No account, no sign-up, no identification of any kind required to begin. [L§3]
- Interface and prompts work in Arabic, including right-to-left reading. [L§12]
- Short by design: one note, one share, then the product encourages the user out. There is no open-ended chat surface to drift into. [Team]
- Neutral wording throughout ("stress", "pressure", "exams"), never clinical labels. [L§3]

## 7. What the Product Observes and Records

Place on the chain: **S4 Observation**. Serves N3, N4, N6. Addresses P2, P4.

### 7.1 What the product receives

Only what the user deliberately gives it inside the product:

| Information                                                | Why the product needs it                                                                                               |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Chip selections                                            | To shape the drafts, to offer a same-session invitation to write, and (if the user allows) to feed the private record. |
| The few lines the user types                               | To produce the drafts for that note.                                                                                   |
| Recipient choice and tone choice                           | To adapt the drafts to the audience.                                                                                   |
| Whether the user accepts or dismisses a writing invitation | To avoid repeating an invitation the user has declined (see §9).                                                       |

### 7.2 What the product does not observe

To keep the product free of any surveillance reading, this list is a requirement, not a courtesy:

- Nothing outside the product: no messages, contacts, browsing, location, microphone, camera, or other apps.
- No typing patterns, device sensors, or behavioural signals used to infer mood or condition.
- No scoring, screening, symptom labels, or inference of any condition. [L§3]
- No tracking of whether, when, or with whom a note was shared.
- No profile of the user, and no record of the user's identity.

### 7.3 The purpose of observation

The purpose is narrow: to support the user's own experience — the drafts they get now and the quiet invitation they may receive later. It is not to monitor, rate, or report on the user. Nothing observed is shown to anyone other than the user.

This is also how the product meets the early-intervention goal [L§6] without becoming a detection or diagnosis tool: it does not "detect distress"; it notices only that the user keeps choosing the same situation, and offers help writing about it.

## 8. Safe On-Device Record

*Place on the chain: **S5 Private on-device record**. Serves N3, N4, N6. Addresses P2, P4.*

### 8.1 What it is

A small private memory, on the user's own device, of the user's recent chip selections. It makes recurring pressure visible to the product so it can offer a timely, relevant question. It is not a general notes feature, and it is not required to use the product.

### 8.2 Product requirements for the record

1. **User-controlled.** The record exists only with the user's knowledge and choice. **Assumption:** it is off until the user turns it on. **Decision requiring validation:** the default state (see Q6).
2. **On-device only.** It is never synced, never backed up elsewhere, never transmitted, and never sent in any form — not even summarised or scrubbed — to any service. The drafting process never sees it. [L§3, Team]
3. **Minimum content.** It holds only what is needed to recognise a recurring pressure: which situation chips were selected and roughly when. **Proposal:** it does not hold free text, drafts, recipient identities, or anything about whether a note was shared. **Unresolved question:** whether the user may also choose to keep written notes privately (see Q7).
4. **Protected.** Someone who picks up or borrows the phone should not be able to casually read it. The strength and method of protection is an implementation choice (§16, choice IC5); the principle is not negotiable. [L§3]
5. **Visible and deletable.** The user can see what is remembered and can erase all of it in one action at any time.
6. **Time-limited.** Entries expire after a short retention window that the user can set. **Unresolved question:** the default window (see Q9).
7. **Purpose-limited.** It is used only to decide whether to offer a writing invitation. It is never used to score, label, or characterise the user.
8. **Optional by design.** If the record is off, empty, or unavailable, the product still works end to end. The same-session invitation (§9) still functions.

### 8.3 Why on-device is the product position

A saved history of someone's emotional pressures is the most sensitive information this product could ever hold. The challenge explicitly asks teams to avoid collecting unnecessary sensitive data [L§3]. Keeping the record only on the user's device means there is no stored collection anywhere that could be breached, requested, or misused; the user keeps the only copy, and deleting it is fully in their hands. The product still

delivers the moment of being noticed — it just does so without anyone else holding the information.

## 9. The Writing Trigger

*Place on the chain: **S6 Writing trigger**. Serves N1, N6. Addresses P1, P2.*

### 9.1 What it is

An appropriate, gentle question that invites the user to begin writing. It is the connection between what the user has been tapping and the opportunity to write.

### 9.2 Two kinds of trigger

| Kind                 | When                                                          | Needs the private record?   | Example                                                                                            |
|----------------------|---------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------|
| Same-session trigger | The user has just selected one or more chips.                 | No                          | "You picked <i>exams</i> . Want help writing a short note to someone about it?"                    |
| Pattern trigger      | The same theme has come up repeatedly within a recent period. | Yes (if the user allows it) | "You've mentioned exams a lot lately. Do you want help writing a short note to a friend about it?" |

The same-session trigger means the product's central behaviour — interaction leading to an invitation to write — works even when no history exists or the record is switched off.

### 9.3 Product requirements for the trigger

1. **Optional and easy to decline.** One tap to dismiss. Never blocks the product. Never repeated immediately after a refusal.
2. **Neutral and non-clinical.** It names the situation the user chose ("exams"), never a feeling, condition or judgement about the user. It never says or implies the user "seems" anything.
3. **About writing, not about the person.** It offers help writing a note; it does not assess, advise, or probe for distress. [L\$3, Team]
4. **Quiet.** Not frequent, not alarming, not attention-seeking. The product's aim is to get the user out of the product and into a conversation, not to bring them back.  
**Unresolved question:** how often and after what recurrence the pattern trigger should appear (see Q8).

5. **Reviewed wording.** The set of questions is small, written and reviewed by a native Arabic speaker, with one per theme. The question's wording is therefore predictable. **Needs research:** how the wording lands with real young people, and whether it feels helpful or intrusive.
6. **Local only.** The decision to show a trigger depends only on information that stays on the device.
7. **A trigger never acts.** It never contacts anyone, never sends anything, and never moves information off the device.
8. **Mechanism is open.** The exact way the question is chosen is an implementation choice (§16, choice IC6). A simple, predictable mechanism is acceptable and may be preferable.

## 10. The Writing Experience

*Place on the chain: **S7 Writing experience.** Serves N1, N2, N4, N5. Addresses P1, P3, P4.*

### 10.1 What the user experiences

1. **Begin.** The user accepts a trigger, or opens the product and starts directly. The chips they selected are already shaping what follows.
2. **Write.** One to three lines, in their own words. Chips alone are enough if they cannot write.
3. **Safety check.** Before any drafting, the Guardian Layer checks what the user wrote (§10.4).
4. **Drafts.** Three drafts appear — gentle, direct, formal — each adapted to the chosen recipient, each traceable to the user's words.
5. **See, compare, edit.** The user can see which phrases came from their own words, compare with a plain template, and edit the text. Nothing is final until they approve it.
6. **Choose to share.** The user sends the note through a messaging app they already use, or chooses not to.
7. **Support is visible.** Where to find a human is always on screen.

### 10.2 Drafting requirements

- Exactly three tones (gentle, direct, formal), adapted to the recipient.
- Written from the user's own words only. **No invented facts.** Every generated claim must trace to something the user typed.
- No condition names, no diagnostic language, no medication or treatment suggestions, no clinical advice. [L§3]

- Neutral framing; "I-statements" ("I've been feeling...", "I need...") rather than blame.
- The user can always edit; the product never sends; the product states clearly that AI helped draft the text. [L§3 — transparency]
- Output that violates these rules is checked and, if necessary, replaced with a plain template rather than shown partially.

### 10.3 Trust and control capabilities

- **Faithfulness view:** shows which parts of a draft come from the user's input, so the user can see nothing was invented.
- **Baseline comparison:** shows the AI draft against a plain template, so value is visible.
- **Outbound preview:** before anything leaves the device for drafting, the user can see exactly what would leave, with personal identifiers removed, and can correct it.
- **Persistent human route:** a control present on every screen that always works, regardless of any other state.

### 10.4 The Guardian Layer (mandatory)

The Guardian Layer is a wrapper around the writing experience, not a feature inside the drafting. No drafting happens until it has run. [L§3, TP]

#### Stage 1 — Risk check, before any drafting.

Everything the user types is checked before it can reach the drafting step. The check is layered so one failure does not cause a miss: a team-maintained list of Arabic and Libyan-dialect crisis phrasing reviewed by an Arabic-speaking member, and a model-based judgement run alongside it. Both are **high-recall by design**: the check is tuned to over-trigger rather than under-trigger. A false alarm costs one extra support card; a false negative costs a person.

#### Stage 2 — Fixed support card.

On a trigger the product does not draft. It shows a static, pre-written card with verified human contacts. The card text is never generated, never personalised by a model, never paraphrased, so it cannot drift under pressure. Afterwards the user may continue to the note if they still want to.

#### Stage 3 — Policy and length discipline.

The safety rules are applied at every drafting step, not just once. Sessions are deliberately short: one note, one share, then encouragement out of the product.

#### Stage 4 — Output check.

A final deterministic check on the drafts blocks diagnostic wording, condition names, medication and dosage references, treatment plans, and anything that reads as clinical advice. A blocked draft is regenerated once, then replaced with a plain template.

#### Stage 5 — Evidence.

The team assembles a test set of Arabic crisis phrasings — including Libyan dialect, mixed Arabic/English and euphemistic phrasing — and reports the check's recall and false-alarm rates. A safety claim without a measured number is an assertion, and will be treated as one.

Why the check runs first. If it ran only on output, the drafting step would already have seen the most sensitive text. Running it first means crisis text never enters drafting at all. Combined with the persistent human-route control, the user always has a route to a human that does not depend on the classifier being correct.

Verified contacts — a hard dependency. Every contact on the support card must be verified by a named team member before any demo. Never show an unverified number. Unresolved question: no verified list of Libyan support contacts exists in the project material (see Q1). If none can be verified, the card must say plainly that the product cannot display a contact it has not confirmed, and point the user to a trusted person.

### 10.5 Data flow of the writing experience (product view)

| Stage                                   | What exists                                    | Leaves the device?                     |
|-----------------------------------------|------------------------------------------------|----------------------------------------|
| User selects chips and types the note   | chips + one to three lines                     | — ▾                                    |
| Personal identifiers removed            | placeholders replace names and numbers         | — ▾                                    |
| Prepared text for drafting              | identifier-free text                           | <b>Only this, and only to pro...</b> ▾ |
| Drafts returned                         | three drafts                                   | — ▾                                    |
| User edits and shares                   | final text inside the user's own messaging app | Only when the user shar... ▾           |
| Private record (if the user enabled it) | recent chip selections                         | <b>Never</b> ▾                         |

There is no account, no session history, no stored copy of note contents, and no tracking of whether or with whom a note was shared. Needs research: what any outside drafting service itself retains, and whether that can be turned off (see §18).

## 11. The Intended Outcome

*Place on the chain: **S8 Intended outcome**. Serves N1, N4, N5, N6. Addresses P1, P2, P3.*

### 11.1 The outcome we are trying to achieve

A young person moves from silence to a conversation with someone already in their life — a friend, sibling, parent, teacher or counsellor — early, before the situation becomes severe, and on their own terms. If the situation is more serious, they know exactly where to turn and are never left without a route to a human.

### 11.2 What success looks like

- The user reaches the share step with a note they are willing to send.
- The user feels the note says what they mean, in a tone that suits the recipient.
- The user is routed to a human, never to a longer conversation with the product.
- No one is diagnosed, profiled, or left with data behind.

### 11.3 What the product impact story is

The specialist workforce in Libya is small. The product does not add clinical load. It moves a person from silence to someone already available in their life, and routes to a specialist only when the Guardian Layer or the user chooses that path. [L§2, L§6, L§12]

### 11.4 How the outcome can be evidenced

Because the product deliberately does not track sharing, outcomes cannot be measured in the field. They are evidenced through:

- Demonstration sessions and small consented trials with friends, observing whether participants reach the share step.
- Measured safety figures (recall and false-alarm rates) and faithfulness figures (§18).

**Unresolved question:** how to present outcome evidence honestly to the committee without tracking users (see Q15).

## 12. Safety and Privacy Considerations

*These apply at every stage of the chain. Each is also stated where it arises (§6 to §11).*

### 12.1 Principles

| Principle                              | What it means                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No diagnosis</b>                    | The product never names or predicts a condition, and produces no scores, screens or symptom labels. [L§3]                                                                                                                                                 |
| <b>No replacement of professionals</b> | Every flow ends in a human. The product states that it is not a specialist and not a crisis service; it is a route to one. [L§3]                                                                                                                          |
| <b>Nothing is automatic</b>            | The product never contacts a third party — friend, parent, hotline or authority — without the user's explicit action. In a culture where mental health disclosure is a family matter, this is the single most important design decision for trust. [Team] |
| <b>Minimum data</b>                    | The least information needed, in the fewest places, for the shortest time. [L§3]                                                                                                                                                                          |
| <b>Private record stays private</b>    | The record of chip selections is potentially personal. It is on-device, user-controlled, never transmitted, and used only for the user's own writing invitation. [L§3]                                                                                    |
| <b>Safe before helpful</b>             | Risk is checked before drafting; the human route never depends on the check.                                                                                                                                                                              |
| <b>Transparency about AI</b>           | The product states that it is an AI at every drafting step, and shows what it did. [L§3]                                                                                                                                                                  |
| <b>Honest limits</b>                   | The product and the report state what it cannot do.                                                                                                                                                                                                       |

### 12.2 Stated limits

**What Bridge Note is not:** a doctor; a diagnosis tool; a replacement for specialists; a crisis service (it is a route to one); a therapist (it does no therapy or exercises).

Known limitations to state openly:

- 1. The AI can still produce an unfaithful draft. Mitigation: the faithfulness view, editability, and the fact that the AI never sends.
- 2. The risk check can miss a crisis phrasing, especially unusual dialect or heavy slang. Mitigation: layered checks, high-recall tuning, a human-route control that does not depend on the check, and published recall numbers rather than a claim of completeness.
- 3. Identifier removal can miss a misspelled name. Mitigation: the outbound preview lets the user be the final check.
- 4. The safety phrase list is not exhaustive. Mitigation: native review, ongoing maintenance, and improvement from the test set.
- 5. Whoever picks up the user's phone may see the product. Mitigation: protected record, one-action delete, and neutral naming.
- 6. Any outside drafting service may have its own handling of text. Mitigation: disclosed, minimised, and researched (§18).

12.3 Minors and consent

The target users are young people, and some are under 18. The product creates no account and collects no identifying data, which removes most consent surface.

**Working position:** no data collection, no account, no third party contacted, and the user decides who receives a note, so the product never creates a disclosure the user did not choose. **Unresolved question:** how a minor uses the product, and what the product says about age and consent. This must be stated in the interface and in the report (see Q5).

PART III — WHAT THE PRODUCT MUST ACCOMPLISH

13. Product Requirements

Requirements say **what** the product must accomplish. None names a technology.

| ID | Requirement                                           | Serves | Stage | Source         |
|----|-------------------------------------------------------|--------|-------|----------------|
| R1 | The user can begin with chip selections alone and get | N1     | S3    | [L§12, Team] ▾ |

| ID | Requirement                                                                                                                                                                        | Serves     | Stage | Source   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|----------|
|    | something useful without typing.                                                                                                                                                   |            |       |          |
| R2 | The user can write one to three lines in Arabic, Libyan dialect, mixed Arabic/English or Latin-letter Arabic.                                                                      | N1, N7     | S3    | [L§12] ▾ |
| R3 | The user chooses the recipient from a set including friend, sibling, parent, trusted adult, and counsellor/teacher.                                                                | N4         | S3    | [Team] ▾ |
| R4 | The product receives only information the user deliberately provides inside it, and observes nothing else.                                                                         | N3         | S4    | [L§3] ▾  |
| R5 | The product can keep a private record of the user's own recent chip selections on the user's device only, under the user's control (see; delete; time-limited), never transmitted. | N3, N4, N6 | S5    | [L§3] ▾  |
| R6 | The product works fully with the record off, empty or unavailable.                                                                                                                 | N1         | S5–S7 | [Team] ▾ |
| R7 | The product can offer a gentle, optional, neutral invitation to write — from the current session's chips and, where the user allows, from recurring chips.                         | N6         | S6    | [L§6] ▾  |
| R8 | An invitation is easy to decline, never repeated immediately after a refusal, never clinical, and never acts.                                                                      | N2, N4     | S6    | [Team] ▾ |

| ID  | Requirement                                                                                                                                                | Serves | Stage | Source       |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|--------------|
| R9  | Before any drafting, the user's text is checked for risk, tuned to over-trigger rather than miss.                                                          | N5     | S7    | [L§3] ▾      |
| R10 | On a risk detection the product shows a fixed, pre-written, verified support card instead of drafting, and lets the user continue afterwards if they wish. | N5     | S7    | [L§1, L§3] ▾ |
| R11 | A human route is reachable on every screen at all times, independent of any automated judgement.                                                           | N5     | All   | [L§3, TP] ▾  |
| R12 | The product produces three drafts (gentle, direct, formal) adapted to the recipient, from the user's own words only.                                       | N1, N2 | S7    | [Team] ▾     |
| R13 | Drafts contain no invented facts, no diagnosis or condition names, no medication or treatment suggestions, no clinical advice.                             | N2     | S7    | [L§3] ▾      |
| R14 | The user can edit any draft; nothing is final until the user approves it.                                                                                  | N4     | S7    | [Team] ▾     |
| R15 | The product never sends anything and never contacts anyone on the user's behalf.                                                                           | N4     | S7–S8 | [Team] ▾     |
| R16 | What leaves the device for drafting is limited to what is needed for the current note, with personal identifiers                                           | N3     | S7    | [L§3] ▾      |

| ID  | Requirement                                                                                                                                                   | Serves | Stage | Source        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|---------------|
|     | removed, and the user can see it first.                                                                                                                       |        |       |               |
| R17 | Nothing from the user's notes is kept after the session, and no account or identifier is required.                                                            | N3     | S7–S8 | [L§3] ▾       |
| R18 | The product states that it is an AI and shows what it did (faithfulness view; comparison with a plain template).                                              | —      | S7    | [L§3, L§4] ▾  |
| R19 | Sessions are short: one note, one share, then encouragement out. No open-ended chat.                                                                          | N5     | S7    | [Team] ▾      |
| R20 | Every support contact the product displays has been verified by a named person beforehand.                                                                    | N5     | S7    | [Team] ▾      |
| R21 | The interface is Arabic-ready (including right-to-left), tolerant of patchy connections, and usable on an ordinary phone.                                     | N7     | All   | [L§12] ▾      |
| R22 | Safety claims are backed by measured figures (risk-check recall and false-alarm rates; faithfulness; blocked-output tests), reported with their sample sizes. | N5     | S7    | [L§3, L§12] ▾ |
| R23 | The product states plainly what it is not (not a doctor, not a diagnosis tool, not a replacement for specialists,                                             | —      | All   | [L§3] ▾       |

| ID  | Requirement                                          | Serves | Stage | Source            |
|-----|------------------------------------------------------|--------|-------|-------------------|
|     | not a therapist, not a crisis service).              |        |       |                   |
| R24 | A working prototype that can be demonstrated online. | —      | S1–S8 | [L§7, L§8, L§9] ▾ |

## 14. Required Capabilities

Capabilities are the product behaviours that satisfy the requirements. Each has a visible place in the chain.

| ID | Capability                                                              | Satisfies         | Stage | Problem / need served | Contributes to outcome by...                                 |
|----|-------------------------------------------------------------------------|-------------------|-------|-----------------------|--------------------------------------------------------------|
| C1 | Chip capture with the recipient selector                                | R1, R3            | S3    | P1 / N1 ▾             | making the first step possible with one tap                  |
| C2 | Short free-text writing in the user's language and script               | R2                | S3    | P1 / N1, N7 ▾         | letting the user say it in their own words                   |
| C3 | Bounded observation (only what the user provides)                       | R4                | S4    | P4 / N3 ▾             | keeping the product free of surveillance                     |
| C4 | Private on-device record of chip selections                             | R5, R6            | S5    | P2 / N6, N3 ▾         | giving the product memory without anyone else holding data   |
| C5 | Writing trigger (same-session and pattern)                              | R7, R8            | S6    | P1, P2 / N6 ▾         | turning a recurring pressure into an invitation to write     |
| C6 | Guardian Layer (risk check, fixed support card, output check, evidence) | R9, R10, R13, R22 | S7    | P4 / N5 ▾             | making sure help is a human, and never harm                  |
| C7 | Persistent human route                                                  | R11, R20          | All   | P3, P4 / N5 ▾         | guaranteeing a path to a person that never depends on the AI |
| C8 | Drafting engine (structure, three tones, fidelity)                      | R12, R13, R14     | S7    | P1 / N1, N2 ▾         | producing a message the user is willing to send              |

| ID  | Capability                                                                         | Satisfies | Stage | Problem / need served | Contributes to outcome by...                                 |
|-----|------------------------------------------------------------------------------------|-----------|-------|-----------------------|--------------------------------------------------------------|
| C9  | Trust and control views (faithfulness view, baseline comparison, outbound preview) | R16, R18  | S7    | P4 / N3, N4 ▾         | letting the user and a judge see the AI is honest and useful |
| C10 | User-controlled sharing through the user's own messaging apps                      | R15, R14  | S7–S8 | P1, P3 / N4 ▾         | ending the interaction in a real human conversation          |
| C11 | Data minimisation (identifier removal, nothing retained, no account)               | R16, R17  | S7–S8 | P4 / N3 ▾             | leaving no data behind                                       |
| C12 | Language and context support (Arabic, RTL, dialect, low bandwidth)                 | R2, R21   | All   | P1, P3 / N7 ▾         | making the product usable in Libya                           |
| C13 | Honest limits (stated limits, AI disclosure)                                       | R18, R23  | All   | P4 / — ▾              | earning and keeping trust                                    |
| C14 | Demonstrable prototype                                                             | R24       | All   | — ▾                   | satisfying the challenge's demonstration expectations        |

14.1 Traceability chain

Every capability can be traced along the same chain: Problem (§2.2) → User need (§3.1) → Product idea (§4) → Requirement (§13) → Capability (§14)

→ Behaviour (§6–§10) → Chain stage (§5) → Outcome (§11)

Example: **P2** (early signs go unshared) → **N6** (to be gently noticed) → idea: private record + invitation → R5, R7, R8 → C4, C5 → behaviour in §8 and §9 → stages S5–S6 → outcome: the user begins a note about a recurring pressure early.

Example: P4 (risk of harm) → **N5** (reliable route to a human) → **idea**: Guardian Layer + persistent human route → **R9, R10, R11, R20** → **C6, C7** → §10.4 → stage **S7** → outcome: the user is never left without a route to a person.

## 15. Alignment With the Challenge Requirements and Evaluation Criteria

### 15.1 What the solution must consider [L§3]

| Challenge requirement                                               | How the product satisfies it                                                                                                                         | Where                   |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Privacy</b> — avoid collecting unnecessary sensitive data        | No account; drafting path keeps nothing; the private record is optional, on-device and user-controlled.                                              | §7, §8, R4–R5, R16–R17  |
| <b>Data security</b> — reduce risk of unauthorised access or misuse | Minimum information leaves the device; identifiers removed first; no stored copy of notes anywhere to breach; the record is protected on the device. | §8, §10.5, C11          |
| <b>AI ethics</b> — responsible, transparent, safe                   | Disclosure that AI is used; faithfulness view; baseline comparison; the user approves every word.                                                    | §10.3, R18              |
| <b>Safety</b> — no psychological harm or dangerous guidance         | Guardian Layer; fixed support card; output check; persistent human route; short sessions.                                                            | §10.4, R9–R11, R13, R19 |
| <b>No medical diagnosis</b>                                         | Generation constraints and output check forbid conditions and diagnostic language; no scores or screens.                                             | §7.2, R13               |
| <b>Do not replace professionals</b>                                 | Every flow ends in a human; the AI never contacts anyone; stated limits.                                                                             | §12, R15, R23           |

## 15.2 Evaluation criteria [L§4]

| Criterion                                                         | How the product answers it                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Real use of AI</b> ( <i>one of the two most important</i> )    | Four concrete AI functions (§4.4): free-text structuring, tone adaptation, faithfulness checking, risk classification. Demonstrable live through the faithfulness view and baseline comparison.                                                                                                                                                                                                                                                                                                     |
| <b>Applicability in Libya</b> ( <i>the other most important</i> ) | Arabic and right-to-left; Libyan dialect and mixed-script input; low-bandwidth tolerance; no account; ordinary phones; the user's own messaging apps; verified local contacts; respect for family and community in the recipient choices (see §15.3).                                                                                                                                                                                                                                               |
| <b>Innovation</b>                                                 | Related work exists in AI-assisted supportive messaging and in AI-guided conversation practice. The new element here is the <b>direction</b> : the AI helps the help-seeker reach a person, rather than talking to the user or helping a supporter. The second new element is the connected loop from the user's own taps, to a private on-device memory, to a quiet invitation to write.<br><b>Assumption</b> : no identical product is known; to be stated as "we found none", not "none exists". |
| <b>Social impact</b>                                              | Targets the silence that delays help-seeking; early intervention is the lecture's stated goal; adds no load to scarce specialists.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Ease of use</b>                                                | Chips remove the blank page. Useful with zero typing, better with three lines. Three tones, one tap to share.                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Privacy and safety</b>                                         | Stateless drafting; on-device private record; identifier removal with preview; Guardian Layer with fixed support card; published safety numbers.                                                                                                                                                                                                                                                                                                                                                    |

## 15.3 Why this works in Libya

"Applicability in Libya" is not satisfied by translating the interface. The product must survive Libyan conditions.

| Libyan reality                                                                           | What the product does about it                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Users write in <b>Libyan Arabic</b> , frequently mixed with English or in Latin letters. | Drafting and risk check are tested on dialect and mixed-script input, not standard Arabic alone. A native reviewer checks all Arabic wording. <b>Needs research:</b> actual model quality (§18).                                                                                                                           |
| <b>Mental health carries stigma</b> ; naming a condition is socially costly.             | Neutral framing throughout; no clinical labels or diagnosis-shaped output; nothing the user shares can read as a label they attached to themselves.                                                                                                                                                                        |
| <b>Connections are patchy and mobile data is expensive.</b>                              | The record and trigger work with no connection; the note is short by design; a short drafting request only.                                                                                                                                                                                                                |
| <b>People use common messaging apps</b> , not email.                                     | Sharing goes through messaging apps the user already has. <b>Assumption:</b> WhatsApp and Messenger are the leading candidates; which to prioritise is to be validated (Messenger reaches roughly 94% of the population by one ad-audience estimate from September 2026; comparable WhatsApp data has not been confirmed). |
| <b>Users do not want to be identified</b> where mental health is a family matter.        | No account, phone number, email or nickname; anonymous by default.                                                                                                                                                                                                                                                         |
| <b>Not everyone has a recent phone or spare storage.</b>                                 | Small footprint; the optional record is the only persistence.                                                                                                                                                                                                                                                              |
| <b>Verified human support is scarce but exists.</b>                                      | The support card lists only verified contacts; the product's whole purpose is to route users toward those scarce humans rather than substitute for them.                                                                                                                                                                   |

| Libyan reality                                                                             | What the product does about it                                                     |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Users may prefer support that respects their own values</b> (family, faith, community). | The recipient set includes trusted family and a trusted adult; nothing is imposed. |

## 15.4 What the evaluators want to see [L§12]

| Evaluators want                                                   | Where it is answered |
|-------------------------------------------------------------------|----------------------|
| A clear problem related to stress among young people in Libya     | §2                   |
| A specific solution, not only a broad idea                        | §4, §10              |
| A clear and meaningful role for AI                                | §4.4                 |
| A practical and safe approach to protecting user data             | §7, §8, §10.5, §12   |
| Early support or guidance without diagnosis                       | §9, §10.4, §12       |
| Clear social value and realistic applicability in Libya           | §11, §15.3           |
| A short, convincing presentation, ideally with a simple prototype | §20, §22             |

# PART IV — HOW IMPLEMENTATION DECISIONS WILL BE MADE

## 16. Implementation Choices to Be Made

Nothing in this part is a final decision. The process for each choice is: product requirement → possible approaches → research → criticism / validation

→ evaluation against project constraints → selection → implementation

For each choice below: the capability to be solved, reasonable approaches, what the choice must satisfy, relevant constraints, and what needs research or validation. Approaches are listed to be evaluated, not to be adopted by default.

## IC1 — The form in which the product is delivered (*capability: the whole product, C14*)

- **Approaches:** an installable phone application; a web application used on a phone; another lightweight form.
- **Must satisfy:** works on ordinary phones (R21); demonstrable online to evaluators [L§9, L§10]; tolerates poor connections; no account.
- **Constraints:** four people and under a day; the team's existing skills; how evaluators will open the demo; whether the record's on-device protection is achievable in the chosen form.
- **Needs research / validation:** how evaluators will access the demo; what each team member can build fastest; what private on-device memory each form supports and how well it can be protected.

## IC2 — How drafting intelligence is provided (*capability: C8, partly C6*)

- **Approaches:** a hosted language-model service called directly by the product; the same service reached through a small intermediary the team controls; an open-source model run on the device; an open-source model run on a server the team controls. Open-source models, APIs and frameworks are permitted if cited [T§4].
- **Must satisfy:** good quality in Arabic and dialect; fast enough for a live demo; nothing retained from notes (R17); a way to keep any access key from being exposed; availability and billing that work from Libya.
- **Constraints:** under a day; four people; demo reliability; cost; the more the product relies on a remote service, the more it depends on the network.
- **Needs research / validation:** quality on Libyan dialect and Latin-letter Arabic using real test inputs; whether the provider is accessible and payable from Libya, and its limits; what the provider retains and whether that can be disabled; whether a device-run model is realistic in the window; what happens when the service is unavailable during a demo.

## IC3 — How the risk check works (*capability: C6*)

- **Approaches:** a phrase list alone; a model-based judgement alone; both layered (the current product intention); additional checks.
- **Must satisfy:** high recall on Arabic and dialect crisis phrasing (R9); quick; the support card never depends on it (R11).
- **Constraints:** the available test material; reviewer availability; latency during the demo; cost.

- **Needs research / validation:** the recall and false-alarm rates on a team-built test set; whether any existing Arabic crisis phrase resources are usable and what their licences allow; how to test dialect and euphemistic phrasing; who reviews.

## IC4 — How personal identifiers are removed before anything leaves the device (*capability: C11*)

- **Approaches:** pattern and word-list rules; an on-device language model; asking the user to review and correct (a preview); combinations.
- **Must satisfy:** removes names, phone numbers, and emails (R16); misses are visible to the user; adds no heavy dependency.
- **Constraints:** under a day; Libyan names and spelling variants; mixed scripts.
- **Needs research / validation:** coverage on realistic Libyan inputs; how well the preview lets the user catch a miss; whether a heavier approach adds real value within the window.

## IC5 — How the private record is kept safely (*capability: C4*)

- **Approaches:** protected on-device storage; held in memory for the current session only (no persistence); other on-device options.
- **Must satisfy:** only on the device (R5); not casually readable by someone who picks up the phone; erasable in one action; excluded from any backup; works when absent (R6).
- **Constraints:** setup effort versus value; how strong the protection must be to be defensible; the chosen delivery form (IC1).
- **Needs research / validation:** what protection level is defensible given what the record contains; what each delivery form offers; the cost of setting it up relative to its contribution to the product.

## IC6 — How the writing trigger is formed (*capability: C5*)

- **Approaches:** a small fixed set of reviewed questions chosen by simple counting of recurring chips; time-based reminders; model-generated questions.
- **Must satisfy:** neutral, non-clinical, predictable wording (R8); fully local; no inference of any condition; quiet.
- **Constraints:** a predictable, auditable mechanism is preferred to a clever one; the effect on the user must be tested, not assumed.
- **Needs research / validation:** the number of repeats and period that feel helpful versus intrusive; how the wording lands; whether a model-generated question adds anything that justifies the risk.

## IC7 — How trust and control views are presented (*capability: C9*)

- **Approaches:** a faithfulness view; a baseline comparison; an outbound preview; or only some of these.
- **Must satisfy:** the user can see what the AI did and what leaves the device (R16, R18); demonstrates real AI value to evaluators.
- **Constraints:** each adds build and test time.
- **Needs research / validation:** whether users and evaluators understand each; which carries the most demonstration value per hour of effort.

## IC8 — How the note is shared (*capability: C10*)

- **Approaches:** the phone's general sharing facility; direct hand-off to a specific messaging app; copy-to-clipboard.
- **Must satisfy:** the user decides; nothing tracked (R15); no account.
- **Constraints:** works on the team's test devices; needs no extra permissions.
- **Needs research / validation:** which messaging apps are used most by the target users; which approach is simplest to make reliable.

## IC9 — How safety and dialect evidence is produced (*capability: C6, C12; requirement R22*)

- **Approaches:** a team-written test set of phrases; adapting licensed public resources; asking native reviewers to write and check phrasings.
- **Must satisfy:** includes dialect, mixed script and euphemism; the figures are real and reported with sample size; no real person's crisis text enters the project.
- **Constraints:** reviewer time; licence conditions.
- **Needs research / validation:** the minimum sample that gives a meaningful figure in the time; who writes and reviews; licence checks on any external resource.

## IC10 — How the demonstration shows the pattern trigger (*capability: C4, C5, C14*)

- **Approaches:** a clearly labelled demonstration history that is not a real user's data; a short walkthrough that accumulates same-session selections; a recorded demonstration.
- **Must satisfy:** honest — any sample history is clearly labelled as such; shows the interaction-to-writing link.
- **Needs research / validation:** which approach is clearest in the five-minute slot.

## 17. Criteria for Selecting Among Them

Each choice will be judged against the same considerations. The purpose is good decisions, not a large scoring table.

| Consideration                                | What it asks                                                                    |
|----------------------------------------------|---------------------------------------------------------------------------------|
| <b>Feasibility within the available time</b> | Can four people build and test it in under a day?                               |
| <b>Suitability for a four-person team</b>    | Can the work be split without people blocking each other?                       |
| <b>Reliability</b>                           | Will it work on the day, on the team's devices, on the demo network?            |
| <b>Simplicity, where simplicity helps</b>    | Is the simpler approach good enough for the product's purpose?                  |
| <b>Integration difficulty</b>                | How hard is it to make this piece work with the others?                         |
| <b>Testing burden</b>                        | How much testing does it need before we can trust it?                           |
| <b>Demonstration value</b>                   | Does it help evaluators see the product's value?                                |
| <b>Risk of failure</b>                       | What happens if it does not work, and is there a fallback?                      |
| <b>Availability of tools and resources</b>   | Do we have what we need, including access and billing from Libya?               |
| <b>Privacy</b>                               | Does it minimise what is collected and where it goes?                           |
| <b>Safety</b>                                | Does it keep the user safe, with a route to a human that does not depend on it? |
| <b>Contribution to the intended outcome</b>  | Does it move the user toward a real conversation with a person?                 |

**Two guards apply to every choice:**

1. **Do not sacrifice the core experience merely to make implementation easier.**  
The connected chain — taps, private memory, gentle invitation, writing, human route — must remain recognisable.
2. **Do not add complexity that gives the user little value.** Sophistication, familiarity, or theoretical optimality are not reasons to choose something.

The target is the strongest realistic outcome our actual team can implement and demonstrate, not a theoretically perfect system.

## 18. Research and Validation

This document does not report completed research on these questions. It lists what must be researched or tested, and what each answer decides. The working method is: propose → research → challenge → compare → validate → select

Research here serves decisions. It is not a literature review.

| #   | What to find out                                                                                                                                                                                                                                                                                      | Decides    | How                                                                                      |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------|
| RV1 | Whether a candidate model service writes acceptable Libyan Arabic, mixed Arabic/English and Latin-letter Arabic. Published benchmarks suggest general models are weaker at dialect generation than at standard Arabic and do not cover Libya specifically, so treat quality as unproven until tested. | IC2, IC3 ▾ | Run a small set of realistic inputs through each candidate; have a native reviewer rate. |
| RV2 | Whether the provider is accessible, payable and within limits from Libya, and what it retains. (The provider's public list of supported regions includes Libya; billing and limits are separate and unconfirmed.)                                                                                     | IC2 ▾      | Test a real key and payment route; read the provider's data-handling terms.              |

| #           | What to find out                                                                                                                | Decides                 | How                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------|
| <b>RV3</b>  | The risk check's recall and false-alarm rates on a dialect-inclusive test set.                                                  | IC3, IC9 ▾              | Build the set; run each candidate approach; record sample sizes.    |
| <b>RV4</b>  | How well identifier removal handles Libyan names, spelling variants and Latin-letter Arabic.                                    | IC4 ▾                   | Test on realistic examples.                                         |
| <b>RV5</b>  | What level of protection for the on-device record is defensible, and what each delivery form offers.                            | IC1, IC5 ▾              | Check what each form supports; weigh against what the record holds. |
| <b>RV6</b>  | How the trigger wording and frequency feel to real young people: helpful, neutral, or intrusive.                                | IC6 ▾                   | Small consented test with friends; native reviewer on wording.      |
| <b>RV7</b>  | Whether people understand the faithfulness view and baseline comparison, and whether they earn their build time.                | IC7 ▾                   | Quick walkthrough with a few testers.                               |
| <b>RV8</b>  | Which messaging apps the target users use most, and how sharing behaves on the team's devices.                                  | IC8 ▾                   | Ask target users; test on devices.                                  |
| <b>RV9</b>  | How comparable services have handled disclosing AI assistance in supportive messages, and what trust or consent problems arose. | Requirements R14, R18 ▾ | Brief review of comparable cases.                                   |
| <b>RV10</b> | How evaluators will access an online demonstration and what that implies for delivery.                                          | IC1, IC10 ▾             | Ask the organisers (the lecture names a coordinator for timing).    |

| #    | What to find out                                                                | Decides    | How                                      |
|------|---------------------------------------------------------------------------------|------------|------------------------------------------|
| RV11 | Whether any external Arabic crisis or dialect resource is licensed for our use. | IC3, IC9 - | Check each licence before any inclusion. |

**Starting evidence available now (not validation):** Libya's specialist services are limited and stigma is strong, so friends and family are usually the first line of support; AI-assisted supportive messages have drawn criticism in a comparable peer-support setting when recipients later learned a machine helped write them, which is why the user must remain the author and the product discloses its AI role.

## 19. Open Questions, Assumptions, and Decisions Requiring Validation

### 19.1 Unresolved questions

| ID | Question                                                                                              | Why it matters                                                       | How to resolve                                                                                                                                                          |
|----|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | <b>Which human-support contacts can be verified?</b> No verified list exists in the project material. | R10, R11, R20 depend on it; an unverified number is worse than none. | Assign a named owner and a deadline; verify each contact directly. If none can be verified, use the card that states the product cannot display an unconfirmed contact. |
| Q2 | What form does the product take for the demo (IC1)?                                                   | Shapes everything built.                                             | RV10, RV5, team skills.                                                                                                                                                 |
| Q3 | Which model service, and is it accessible from Libya (IC2)?                                           | The drafting depends on it.                                          | RV1, RV2.                                                                                                                                                               |
| Q4 | Who is the native Arabic reviewer for wording, phrase list and test set?                              | Dialect and safety quality.                                          | Name a team member or reachable reviewer.                                                                                                                               |
| Q5 | How does a minor use the product, and what does it say about age and consent?                         | Target users include under-18s.                                      | Decide a stated position; reflect it in the interface and report.                                                                                                       |

| ID  | Question                                                                                              | Why it matters                                        | How to resolve                                                                                              |
|-----|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Q6  | Is the private record on or off by default?                                                           | Privacy position and demo behaviour.                  | Working assumption: off until the user turns it on; validate against user testing.                          |
| Q7  | Does the product ever let the user keep written notes privately, or only chip selections?             | Scope of the private record and its sensitivity.      | Working proposal: selections only; revisit only if time and a clear benefit exist.                          |
| Q8  | How many recurrences, over what period, trigger the pattern invitation, and how often may it appear?  | Appropriateness versus intrusion.                     | RV6.                                                                                                        |
| Q9  | What is the default retention period of the record?                                                   | Privacy and usefulness.                               | Decide a short default; the user can change it.                                                             |
| Q10 | Which messaging app is the primary share target?                                                      | Applicability in Libya.                               | RV8.                                                                                                        |
| Q11 | Should repeated recurrence of a pressure also surface the human route, not only a writing invitation? | Early intervention [L§6] versus staying non-clinical. | Decide after RV6, with careful wording.                                                                     |
| Q12 | What is the interface's default language, and is a second language offered?                           | Requirement R21.                                      | Not stated in the project material; decide.                                                                 |
| Q13 | Who is the Team Lead, and whose emails are submitted?                                                 | Required by the terms [T§1, T§2].                     | Team to confirm.                                                                                            |
| Q14 | What is the confirmed presentation time?                                                              | Preparation.                                          | The lecture says Thursday 8 October after the noon prayer; an organiser will confirm the exact time [L§10]. |
| Q15 | How is the outcome evidenced to the committee when the product deliberately does not track sharing?   | §11.4.                                                | Use consented demonstration sessions and measured safety figures.                                           |

| ID  | Question                                                                            | Why it matters | How to resolve             |
|-----|-------------------------------------------------------------------------------------|----------------|----------------------------|
| Q16 | Which of the faithfulness view, baseline comparison and outbound preview are built? | Time (IC7).    | RV7; see cut order in §20. |

## 19.2 Assumptions

| ID | Assumption                                                                    |
|----|-------------------------------------------------------------------------------|
| A1 | Users have a phone and at least one messaging app.                            |
| A2 | Users can write one to three lines in the supported languages and scripts.    |
| A3 | Evaluators will assess the product online. [L§9, L§10]                        |
| A4 | At least one capable model service is reachable from Libya.                   |
| A5 | The team has, or can reach, a native Arabic reviewer.                         |
| A6 | A simple, predictable method for choosing the trigger question is sufficient. |

## 19.3 Decisions requiring validation

- The default state of the private record (Q6) and its retention window (Q9).
- The trigger's recurrence rule and frequency (Q8).
- The delivery form and drafting approach (IC1, IC2).
- Which trust views to build (Q16).

# 20. Practical Scope for the Implementation Window

**The constraint.** Four team members. Less than one day. The submission deadline is 11:50 PM on Wednesday 7 October 2026 [T§6]. A finished, production-grade product

is not required [L§8]; a working prototype or practical demonstration is a strong advantage [L§7, L§8].

One problem, one core user experience [TP]. The core experience is the chain in §5: taps → (private memory) → gentle invitation → writing → human conversation.

## 20.1 Tiers

| Tier              | What it contains                                                                                                                                                                                                                                                                                                                                         | Rule                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core — must exist | Chip capture and recipient choice; short free-text writing; three-tone drafts from the user's own words, editable; the Guardian Layer (risk check before drafting, fixed support card, output check); the persistent human route; the same-session writing trigger; sharing under the user's control; nothing retained; stated limits and AI disclosure. | Never cut. The Guardian Layer and the human route are never cut.                                                                                                    |
| Core extension    | The private on-device record and the pattern trigger.                                                                                                                                                                                                                                                                                                    | Built if the core is stable. If time is short, the record is simplified and the product still demonstrates interaction-to-writing through the same-session trigger. |
| Trust views       | Baseline comparison; faithfulness view; outbound preview.                                                                                                                                                                                                                                                                                                | Built in order of demonstration value per hour.                                                                                                                     |
| Optional          | Private written notes; extra tones or recipients; extra languages.                                                                                                                                                                                                                                                                                       | First to be cut.                                                                                                                                                    |

## 20.2 Cut order when behind

Optional items first; then the pattern record (leaving the same-session trigger); then the baseline comparison; then the faithfulness view. The Guardian Layer, the human route, and the privacy of the drafting path are never cut.

## 20.3 Working split

Four people should not edit the same area. Work can be divided by capability area: Capture and trigger (C1, C2, C4, C5), **Drafting** (C8, C10), **Guardian and evidence** (C6, C7, plus the test set), and Trust, documentation and presentation (C9, C13, the report, README, deck or video, tool citations). The split is to be agreed at the start.

## 20.4 Fixed overhead

Regardless of the approach, the following must be produced and are not optional: the repository, README, detailed technical report in Markdown, a pitch deck or demo video inside the repository, and the citation of every tool, model and service used [T\$4, T\$5]. Time for these must be reserved from the start.

# 21. Product-Level Risks

| Risk                                                                  | Likelihood | Impact                      | Response                                                                                                                             |
|-----------------------------------------------------------------------|------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Support contacts are unverified by demo time                          | Medium ▾   | Severe — showing an un... ▾ | Name an owner and deadline now; if none can be verified, use the card that states the product cannot display an unconfirmed contact. |
| The risk check misses dialect crisis phrasing                         | Medium ▾   | Severe ▾                    | High-recall tuning, layered checks, persistent human route, published recall figures, native reviewer.                               |
| The service used for drafting fails or is unreachable during the demo | Medium ▾   | High ▾                      | Recorded demonstration as a fallback; test early; have an alternative approach ready (IC2).                                          |
| Dialect and mixed-script quality is poor                              | Medium ▾   | High ▾                      | Test early (RV1); light wording; native review; state limits.                                                                        |

| Risk                                                                     | Likelihood | Impact   | Response                                                                                                                      |
|--------------------------------------------------------------------------|------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| The writing trigger feels intrusive or labelling                         | Medium ▾   | High ▾   | Neutral wording; optional; easy to decline; quiet; small test (RV6).                                                          |
| The private record is seen as surveillance, or exposed on a shared phone | Medium ▾   | High ▾   | On-device only; user-controlled; visible; one-action delete; purpose-limited; clear statement of what is not observed (§7.2). |
| A draft is unfaithful to the user's words                                | Medium ▾   | Medium ▾ | Faithfulness view; editing; the AI never sends.                                                                               |
| Evaluators read "AI" as decorative                                       | Medium ▾   | High ▾   | Name the four AI functions (§4.4) and show one live (baseline comparison).                                                    |
| The product is mistaken for a clinical tool                              | Low ▾      | Severe ▾ | Stated limits in the interface and pitch; output check blocks clinical wording.                                               |
| Time runs out before the connected chain is demonstrable                 | High ▾     | High ▾   | Tiers and cut order (§20); same-session trigger as the guaranteed link.                                                       |
| A minor uses the product without an agreed position on age and consent   | Medium ▾   | Medium ▾ | Decide and state the position (Q5).                                                                                           |
| Outcome claims cannot be evidenced because sharing is not tracked        | Medium ▾   | Medium ▾ | Consented demonstration sessions; measured safety figures (Q15).                                                              |

## PART V — PROJECT OBLIGATIONS

### 22. Submission and Presentation Obligations

Only obligations stated in the project material are listed.

## 22.1 Submission [T§5, T§6]

Submit a GitHub repository link before the deadline: 11:50 PM, Wednesday 7 October 2026. The repository must contain:

| # | Required artifact                                                                                                                                 | Connection to this document                                                                                        |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 | The complete source code.                                                                                                                         | Built from the requirements (§13) and capabilities (§14), using choices made through Part IV.                      |
| 2 | A <b>README.md</b> outlining the project architecture, operational requirements, setup steps and usage instructions.                              | Architecture is decided after validation (Part IV); usage follows the user flow (§5, §10).                         |
| 3 | A detailed technical report in Markdown ( <b>.md</b> ).                                                                                           | Draws on this document's rationale, safety, privacy and evidence (§12, §18). The report is a separate deliverable. |
| 4 | A presentation file or link (pitch deck), <b>or</b> a project demonstration video, with the file or link included directly inside the repository. | See §22.2.                                                                                                         |

**Tools and citations [T§4].** The use of open-source models, APIs and modern frameworks is permitted, provided that all tools, models and services used are explicitly cited in the project documentation. The citation list must be kept current as choices are made.

Never in the repository: secret keys or credentials; any real user data or real crisis text collected from a person; any unverified support contact; any external resource whose licence has not been checked.

**Not required:** a finished production product [L§8]; in-person attendance for the evaluation [L§10].

## 22.2 Presentation [L§9, L§10, L§12]

| Item                 | Obligation                                                                                                                                    |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Format</b>        | Online, not face to face.                                                                                                                     |
| <b>When</b>          | Thursday 8 October, after the noon prayer; the exact time is coordinated with the committee, and a named organiser will confirm it.           |
| <b>Length</b>        | About 5 minutes of presentation, followed by roughly 5 minutes of committee questions (about 10 in total).                                    |
| <b>Slides</b>        | Around 5 slides are more than enough. They should explain the idea, the problem, the solution, and how the solution works.                    |
| <b>Demonstration</b> | A working online prototype may be demonstrated. A prototype or practical demonstration is a strong advantage but not an explicit requirement. |

The five slides naturally follow the chain: the problem (S1–S2); the solution (S3–S7); what the AI actually does (§4.4); safety and privacy (§10.4, §12); live demonstration and honest limits (§11, §12.2).

Questions to be ready for (all answered in this document): How is this not just a chatbot? Where does the user's data go? What if the AI writes something harmful? Does this diagnose anyone? Why is this feasible in Libya?

## 22.3 Team and event obligations [T§1–T§3, L§10]

- Appoint a **Team Lead** and notify the organisers through the team's designated group chat.
- Provide the **email address of each contestant**.
- **Attendance:** in-person attendance on the day of the event is preferred; alternatively, a designated representative may attend if the team wins. Expenses are not covered by the organisers.
- **Media:** by participating, each participant authorises photography and video capture during the closing-day activities for media coverage and event promotion.

- **Winners** are contacted on Thursday 8 October 2026. The award event is associated with the Libya AI Forum on 10/10 [L§10].